Mutants Testing Report

Mounia Touil

For this assignment, I tried to run mutation testing on JFreeChart project using PIT tool, Java 11, and Maven. First, I ran **mvn clean**, then **mvn test**, and at the end I run mutation testing using **mvn org.pitest:pitest-maven:mutationCoverage.**

1- All mutants generated = 110,818
2- The mutants killed by the test suite = 36,667

3- Mutation Score = (Killed / Total) × 100

= (36,667 / 110,818) × 100
= 0.3308 × 100
= 33.08% ≈ 33%

4- Covered mutants ≈ Total – No coverage

≈ 110,818 – 36,937
≈ 73,881 (approx covered)

5- The estimated number of live mutants ≈ Covered – Killed

≈ 73,881 – 36,667
≈ 37,214

Pit Test Coverage Report

Project Summary

Number of Classes	Line Coverage		Mutation Coverage	
498	55%	29658/53605	33%	36667/110818

Breakdown by Package

Name	Number of Classes	Line Coverage		Mutation Coverage	
org.jfree.chart	21	49%	1818/3703	26%	1811/6893
org.jfree.chart.annotations	16	58%	808/1395	35%	920/2644
org.jfree.chart.axis	46	59%	3619/6179	33%	4491/13533
org.jfree.chart.block	16	66%	828/1260	41%	1114/2749
org.jfree.chart.demo	3	0%	0/164	0%	0/425
org.jfree.chart.editor	13	0%	0/967	0%	0/1908
org.jfree.chart.encoders	5	0%	0/107	0%	0/132
org.jfree.chart.entity	13	43%	206/484	25%	257/1042
org.jfree.chart.event	9	59%	47/80	7%	6/91
org.jfree.chart.imagemap	5	52%	42/81	52%	116/223
org.jfree.chart.labels	27	59%	629/1065	38%	777/2029
org.jfree.chart.needle	10	31%	123/392	23%	197/865
org.jfree.chart.panel	2	13%	34/269	6%	32/528
org.jfree.chart.plot	46	65%	7093/10895	32%	7120/22136
org.jfree.chart.plot.dial	12	59%	806/1357	36%	922/2535
org.jfree.chart.renderer	11	67%	1113/1651	48%	1581/3284
org.jfree.chart.renderer.category	24	46%	2197/4749	18%	1717/9499
org.jfree.chart.renderer.xy	32	42%	2280/5493	20%	2284/11383
org.jfree.chart.resources	1	100%	3/3	100%	2/2
org.jfree.chart.servlet	3	0%	0/143	0%	0/273
org.jfree.chart.title	10	60%	726/1205	33%	757/2320
org.jfree.chart.urls	9	78%	256/327	62%	475/764
org.jfree.chart.util	15	21%	180/850	16%	312/1966
org.jfree.data	17	79%	906/1143	64%	1671/2620
org.jfree.data.category	4	76%	307/403	63%	590/935
org.jfree.data.contour	2	0%	0/185	0%	0/412
org.jfree.data.function	4	81%	62/77	44%	76/172
org.jfree.data.gantt	5	66%	293/446	49%	468/961
org.jfree.data.general	14	63%	936/1496	52%	1726/3345
org.jfree.data.io	1	0%	0/57	0%	0/126
org.jfree.data.jdbc	3	0%	0/249	0%	0/429
org.jfree.data.json	1	0%	0/70	0%	0/87
org.jfree.data.json.impl	3	0%	0/185	0%	0/424
org.jfree.data.resources	6	17%	3/18	17%	2/12
org.jfree.data.statistics	14	75%	1046/1392	53%	1664/3139
org.jfree.data.time	24	64%	1651/2562	49%	2776/5654
org.jfree.data.time.ohlc	4	68%	105/155	46%	151/325
org.jfree.data.xml	8	0%	0/199	0%	0/295
org.jfree.data.xy	39	72%	1541/2149	57%	2652/4658

Report generated by PIT 1.3.2

From all the test results, the mutation score is 33%, which is not very high. This means that many mutants are not killed, which shows the test suite is not strong enough and cannot catch many of the changes made by PIT.

Does killed + live = covered?

No, killed + live is not the same as covered.

Because covered mutants include more cases, not only killed and live. There are also other types like **Timed_Out** or **Memory_Error** mutants. These mutants are executed by the test, so they are covered, but they are not counted as normal killed or live.

So the correct idea is: Covered = Killed + Live + other cases (like timeout)

That is why killed + live is not exactly equal to covered.

Conclusion & Opinions on the experience:

From this testing, I notice a few interesting things. First, the PIT tool takes almost 2 hours to finish. This shows me why mutation testing is not used all the time in real companies because it is very slow and can be expensive to run.

Second, during the run I see hundreds of **"Minion exited abnormally due to TIMED_OUT"** warnings. At first I think something is wrong but then I understand this is normal. It happens because some mutations change a loop condition and make it run forever. JFreeChart have many loops for drawing charts so this happen a lot.

Third, I notice that line coverage is 55% but the mutation score is only 33%. This is very interesting because it show that even if tests execute the code, it does not mean the tests are checking the output correctly. This makes the code coverage alone not enough to tell us if the test suite is really effective.